

Spin-Glass and Ising Models for Brand Memory and Campaign Measurement

A practical modeling brief for using statistical-physics ideas to predict campaign response, brand behavior, and durable memory formation.

Prepared artifact	PDF technical brief with formulas, sample simulation graphs, use cases, and Python implementation notes
Core analogy	Consumers or brand associations are spins; campaigns are external fields; memory is an attractor in the energy landscape
Primary outputs	Short-term response, memory overlap, frustration, hysteresis, parameter-chaos stability, and KPI probability

Executive Summary

Spin-glass physics gives marketers a language for moving beyond isolated KPI lift. The method represents a market as many interacting binary states. Some states reinforce each other, some conflict, and campaigns act as temporary or persistent fields that push the system toward a desired brand-memory pattern.

- **Ising model:** best used as a synchronization benchmark. It answers: how strongly do brand associations move together?
- **Spin-glass model:** best used for contradiction and heterogeneity. It answers: where do brand meanings compete, fragment, or leak to competitors?
- **Hopfield / Amit memory layer:** stores desired brand patterns as attractors. It answers: does a campaign retrieve the intended memory?
- **Non-equilibrium dynamics:** separates short-term buzz from long-term memory by tracking response, decay, hysteresis, and aging.

Source Concepts Used

The report adapts four source ideas: rolling Ising/spin-glass coupling matrices from correlation and mutual information; spin-glass economics as heterogeneous interacting agents; neural-network memory as stable stored spin patterns; and aging, frustration, and dynamical mean-field concepts from spin-glass theory.

1. Modeling Brand Memory as a Spin System

Let each variable be a consumer state, a brand association, or a memory trace. A spin of +1 means the state is active, endorsed, remembered, or chosen. A spin of -1 means the opposite. A full consumer or market state is the vector s .

```
s_i in {-1, +1}
```

Examples:

s_ad_recall = +1	respondent remembers the ad
s_brand_link = +1	memory is correctly linked to the brand
s_competitor = +1	competitor is salient
s_consider = +1	respondent is in the consideration set

Energy Landscape

The market is represented as an energy landscape. States with lower energy are more likely. Campaigns do not simply add lift; they reshape the landscape by making some combinations easier to reach and others harder to sustain.

$$E(s; t) = -1/2 * \sum_{i \neq j} J_{ij} s_i s_j - \sum_i h_i(t) s_i$$

J_{ij}	pairwise coupling between brand states
$h_i(t)$	external field: media pressure, creative cue, price incentive, distribution, PR
β	inverse market noise, $\beta = 1 / T$
T	temperature: volatility, category turbulence, consumer randomness

Ising vs. Spin Glass

Ising benchmark:

$$J_{ij}^{\text{Ising}} = |\text{corr}(x_i, x_j)|$$

Spin-glass representation:

$$J_{ij}^{\text{SG}} = \text{corr}(x_i, x_j)$$

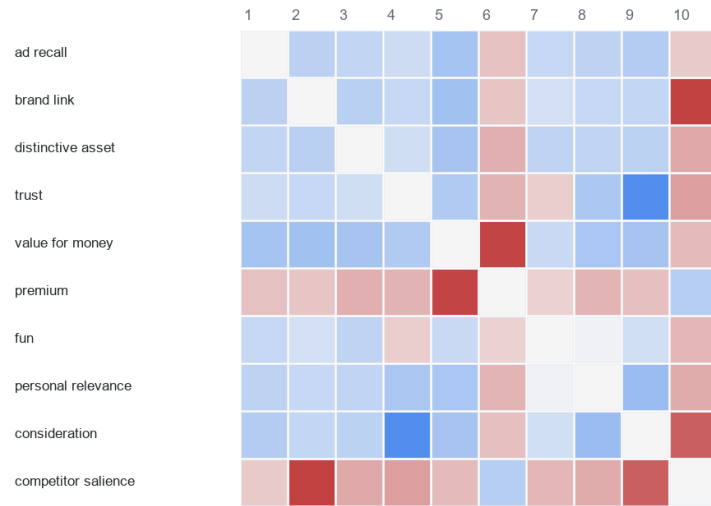
Nonlinear dependence option:

$$\text{MI}(X, Y) = \sum_x \sum_y p(x, y) \log[p(x, y) / (p(x)p(y))]$$
$$J_{ij}^{\text{MI-SG}} = \text{sign}(\text{corr}_{ij}) * \text{normalized_MI}_{ij}$$

The Ising version measures total coherence. The spin-glass version preserves negative relationships: premium vs value, brand link vs competitor salience, or fun vs trust if a category contains that tension.

Signed coupling heatmap: brand associations

Blue = reinforcing association, red = tension or competitive relationship



Numbers across the top map to the same feature order shown on the left.

Figure 1. Example signed coupling matrix estimated from synthetic brand-tracking spins.

2. Campaign as External Field and Memory Attractor

A campaign can be modeled in two ways. First, as an external field $h(t)$ that temporarily increases the probability of selected states. Second, as a memory attractor that strengthens a desired pattern in the coupling matrix.

External Field

$$h_i(t) = h_i^{\text{baseline}} + \text{media_spend}(t) * \text{creative_fit}_i + \text{promo}_i(t)$$

$$P(s_i = +1 \mid s_{-i}) = \text{sigmoid}(2 \beta [h_i(t) + \sum_j J_{ij} s_j])$$

Memory Attractor

The Amit-Gutfreund-Sompolinsky / Hopfield interpretation stores a desired brand-memory pattern x_i as an attractor. If the campaign works, noisy consumer states fall toward that attractor even after the external field fades.

$$J_{ij}^{\text{memory}} = (\alpha / N) * \sum_{\mu} x_{i\mu} x_{j\mu}, \quad i \neq j$$

Memory retrieval / brand-fit score:

$$m_{\mu}(t) = (1/N) * \sum_i x_{i\mu} s_i(t)$$

m_{μ} near +1 desired brand memory retrieved
 m_{μ} near 0 weak or fragmented retrieval
 m_{μ} near -1 opposite or competitor-like pattern

Scenario	Target overlap	Competitor overlap	Consideration	Purchase prob.	Frustration
baseline	-0.168	0.168	-0.227	0.276	0.022
moderate	0.265	-0.265	0.247	0.617	0.022
heavy	0.590	-0.590	0.590	0.825	0.022

Simulated campaign impact on brand-memory state

Synthetic tracker example using signed couplings plus a target memory attractor.

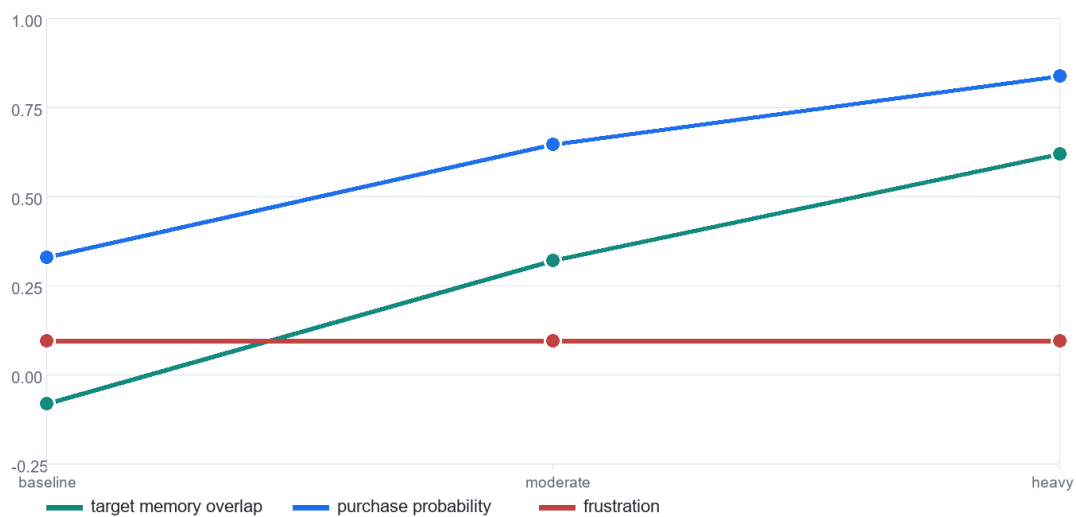


Figure 2. Simulated impact increases target-memory overlap and purchase probability while reducing frustration.

3. Short-Term vs Long-Term Marketing Impact

The short-term/long-term distinction is a non-equilibrium dynamics question. A purely short-term campaign perturbs the market and then decays back to baseline. A long-term campaign changes which states are stable and leaves a residue in memory, preference, or social reinforcement.

Pulse response: fast decay vs memory residue

The same campaign field can produce either temporary response or persistent state change.

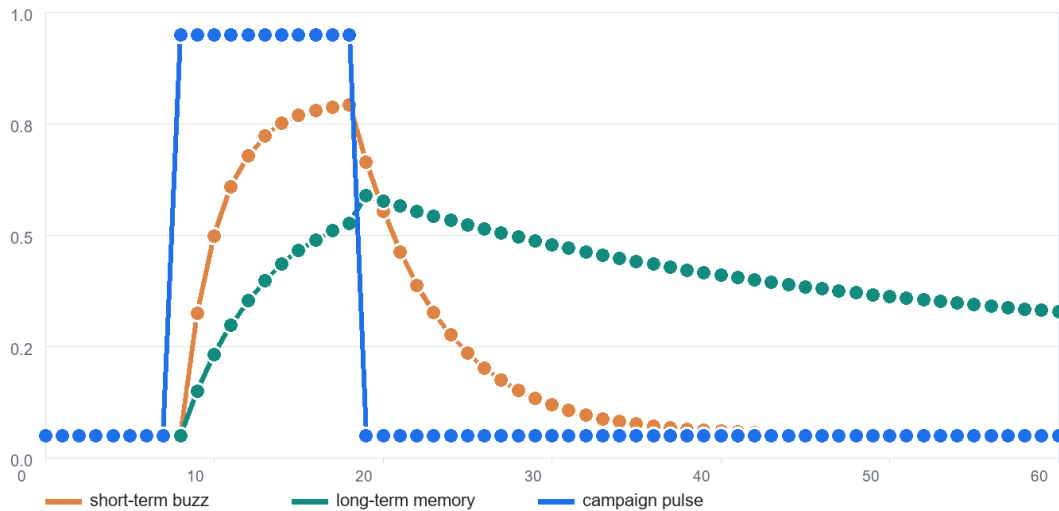


Figure 3. A pulse campaign can generate fast response only, or response plus persistent memory residue.

Short-Term Impact: Fast Regime

Pulse response:

$$R_i(t, t') = \text{partial} \langle s_i(t) \rangle / \text{partial} h_i(t')$$

$$R_M(t, t') = \text{partial} \langle M(t) \rangle / \text{partial} H(t')$$

Aggregate state:

$$M(t) = (1/N) * \text{sum}_i s_i(t)$$

Short-term signature:

$$C(t, t') = C(t - t') \text{ and } R(t, t') = R(t - t')$$

response half-life is short after $h(t')$ is removed

In marketing terms, this is buzz, promo response, search lift, or social chatter that equilibrates quickly. The key metrics are peak response, incremental area under the response curve, decay half-life, and return-to-baseline time.

Long-Term Impact: Slow Regime, Aging, and Memory

Two-time correlation:

$$C(t, t_w) = (1/N) * \text{sum}_i \langle s_i(t) s_i(t_w) \rangle$$

Aging form:

$$C(t, t_w) = C_{\text{fast}}(t - t_w) + C_{\text{slow}}(t / t_w)$$

Long-term signature:

the decay depends on waiting time t_w

the market does not fully return to the old baseline

Long-term marketing impact looks like aging: the system's future response depends on the history of prior exposure. Brand preference, memory structures, distinctive asset linkage, and social proof become part of the slow state variables.

4. Hysteresis, Tipping Points, and Quenched Disorder

Hysteresis: impact depends on path, not only spend level

Increasing and decreasing marketing pressure can produce different market states.

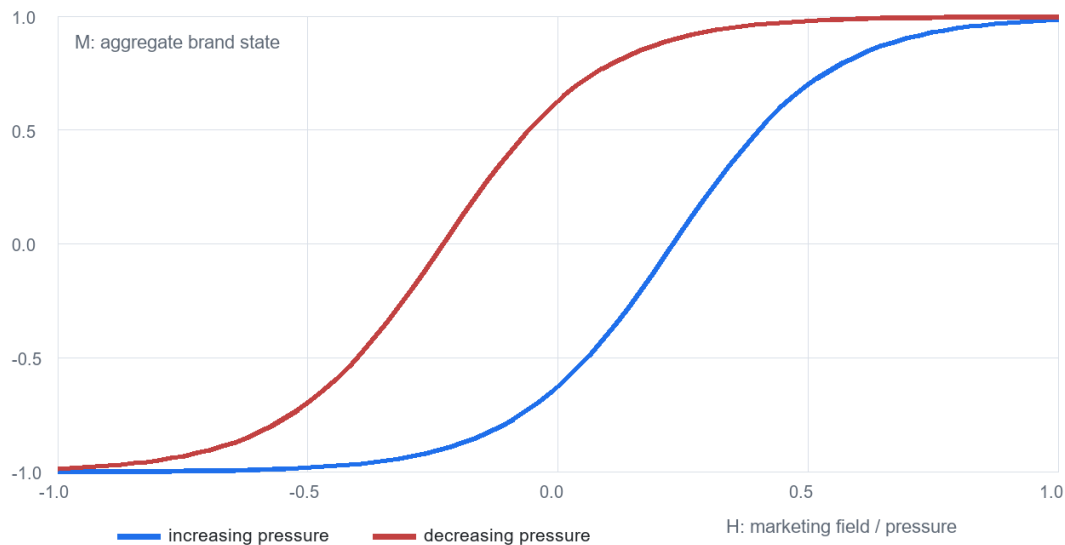


Figure 4. Hysteresis means market state depends on whether pressure is rising or falling.

A Random Field Ising Model adds consumer-level heterogeneity. Some people are easier to move than others, and some resist the campaign even under high pressure.

Random-field brand model:

$$E(s; t) = -\sum_{\langle i,j \rangle} J_{ij} s_i s_j - \sum_i [H(t) + \eta_i] s_i$$

$H(t)$ common marketing pressure

η_i consumer heterogeneity: prior belief, income, need state, category habit

- **Hysteresis:** the brand state under decreasing spend can remain stronger than it was under increasing spend at the same spend level.
- **Tipping point:** once enough consumers or associations switch, social reinforcement and memory couplings make the new state self-sustaining.
- **Quenched disorder:** a successful campaign changes durable structure, not just momentary activation. In practice this appears as changed couplings J_{ij} , stronger brand-link paths, or lower competitor leakage.

Energy landscape: from temporary field to new memory basin

A successful long-term campaign deepens the desired brand-memory minimum.

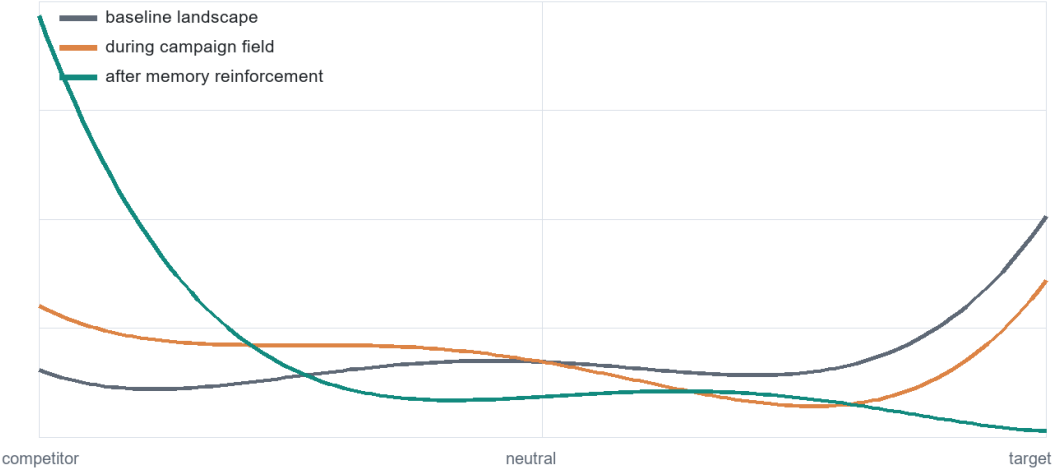


Figure 5. A campaign can tilt the landscape temporarily; durable impact deepens the target memory basin.

5. Measuring Parameter Chaos and Impact Stability

Spin-glass systems can be fragile. A small change in temperature, market noise, macro conditions, or category context can reorder the preferred states. This is useful for measuring whether brand impact will survive a new economic climate.

Overlap under market-condition shift:

$$q(T, T + dT) = (1/N) * \sum_i s_i^{*}(T) s_i^{*}(T + dT)$$

Interpretation:

q near 1	campaign impact is stable across conditions
q near 0	impact is fragile; the market may settle into another state
q below 0	impact reverses or activates a competitor/opposite pattern

Marketing Interpretation

- Stress-test campaigns across inflation, competitor attack, promotion cuts, channel disruption, and category-news shocks.
- Run the model at multiple temperatures T . Higher T means more volatility and weaker predictable memory retrieval.
- Track whether the same target-memory overlap survives or collapses under small parameter shifts.
- Use low-overlap cases to identify fragile assets, overfit creative cues, or segments requiring reinforcement.

Dynamical Mean-Field Approximation

For larger systems, individual interactions can be coarsened into mean fields. This is the practical role of DMFT-style thinking: model the evolving average state without simulating every micro-interaction in full detail.

Mean-field update sketch:

$$m_i(t + 1) = \tanh(\beta [h_i(t) + \sum_j J_{ij} m_j(t) - \text{correction}_i(t)])$$

Campaign question:

Which shocks dissipate, and which shocks move $m(t)$ into a new persistent regime?

6. Applied Use Cases

Use case	Business question	Spin-glass measurement	Decision output
Campaign pretest	Will this creative produce the intended memory?	Targeted memory μ , competitor overlap	Lambda λ (tipping, or adjust media pressure.
Brand-memory tracking	Is the brand becoming more coherent or fragmented?	Building fragmentation, largest eigenvalue λ_1	Find doublets and weakening associations.
Competitive leakage	Does the ad make people remember the competitor?	Overlap with competitor β	Revolve distinctive assets and brand linkage.
Promo vs equity tradeoff	Does short-term sales lift damage long-term value?	Negative memory η value and premium, β	Systemic preference and recovery spend.
Segment strategy	Which audiences are stable, movable?	Segment specific $J, h, q(T, T+dT)$, response	Localize spend to stable conversion or memory-build
Launch tipping point	How much pressure is needed for self-sustaining response?	SFF, Autarky, hysteresis width, basin depth	Estimate minimum viable burst and maintenance spend

Short vs Long-Term Measurement Table

Feature	Short-term impact	Long-term impact
Dynamical regime	Fast regime; rapid equilibration	Slow regime; aging dynamics
Main tool	Response function $R(t, t')$	Two-time correlation $C(t, t_w)$
Landscape effect	Temporary spin shift	New or deeper stable local minimum
Statistical signal	TTI approximately holds	FDT/TTI violations and memory residue
Consumer analogy	Buzz, trend, promo lift	Preference, loyalty, distinctive memory

7. Implementation Blueprint

- **Step 1: Define spins.** Convert survey, search, social, CRM, or purchase variables into +1/-1 states.
- **Step 2: Estimate couplings.** Use signed correlations for spin glass, absolute correlations for Ising, and MI for nonlinear dependence.
- **Step 3: Define the desired memory.** Encode the strategic brand pattern x_i : the associations that should be active or inactive.
- **Step 4: Model the campaign field.** Map media, creative assets, spend, price, distribution, and PR into $h_i(t)$.
- **Step 5: Simulate dynamics.** Use Glauber sampling or mean-field updates under baseline and campaign scenarios.
- **Step 6: Validate externally.** Link overlap, frustration, and response metrics to sales lift, conversion, brand lift, retention, or choice data.

Minimal Python Pattern

```
survey = pd.read_csv("brand_tracker.csv")
spins = binarize_to_spins(survey[FEATURES])

J_sg = corr_couplings(spins, mode="spin_glass", scale=0.55)
J_mi = mi_couplings(spins, signed=True, scale=0.55)

target = np.array([1, 1, 1, 1, 1, -1, 1, 1, 1, -1])
J = J_sg + memory_couplings(target.reshape(1, -1), strength=0.35)

h_campaign = h_baseline + spend * target
samples = glauber_sample(J, h_campaign, beta=1.0)
score = memory_overlap(samples, target)
```

Key Metrics to Report

Metric	Formula / proxy	Marketing meaning
Target memory overlap	$m_{\mu} = (1/N) \sum_i x_i s_i$	Whether the intended brand memory is retrieved
Frustration	share of violated signed edges	Contradiction or fragmented meaning
Pulse response	$R(t,t') = \text{partial } \langle M(t) \rangle / \text{partial } H(t')$	Incremental short-term lift
Memory residue	$M(t+L) - M_{\text{baseline}}$ after field removal	Long-term brand effect
Hysteresis width	H_{up} threshold minus H_{down} threshold	Stickiness of the new state
Parameter-chaos overlap	$q(T, T+dT)$	Stability under market volatility

Cautions

This framework is diagnostic and simulation-based. It should be calibrated with experiments, brand-lift studies, MMM, conversion data, or longitudinal tracking. The model is strongest when used to compare scenarios and uncover structure, not as a stand-alone oracle.

Local Python Asset

The companion prototype in this workspace is `brand_ising_spin_glass.py`. It contains the synthetic simulation, coupling estimators, memory-attractor logic, Glauber sampler, and scenario metrics used to build the sample figures.